



School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Lab 1
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Introduction to Computer Systems and Python. Basic Programming Concepts.

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OBJECTIVES:

- i. To use IDLE to run simple Python commands interactively.
- ii. To code a simple Python program using variables and assignment statements.
- iii. To use the input() function to take user input from the keyboard.
- iv. To use the print() function to display output on the screen.
- v. To save a Python program in a file and then run it using IDLE.

1. Open IDLE on your desktop.

2. Type the following commands one at a time (in the order given below) at the >>> prompt in the IDLE window. A sample screen is shown below. You do not need to submit anything for this step, just make sure you are able to perform the operations given below in IDLE.

- | | |
|------------------------|---|
| a. $15 - 7$ | e. <code>month = 'January'</code> |
| b. <code>z = 20</code> | f. <code>month</code> |
| c. <code>z</code> | g. <code>print('Python is fun!')</code> |
| d. $z / 4$ | |

The Python shell will look as follows:

```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> 15-7
8
>>> z=20
>>> z
20
>>> z/4
5.0
>>> month = 'January'
>>> month
'January'
>>> print('Python is fun!')
Python is fun!
>>>
```

Fig. 1: The results of executing tasks a-g in the Python shell.

3. Create a new Python file by clicking File → New File from the top level menu bar on your IDLE interactive shell. In this new file, write a Python program with comments to do the following.

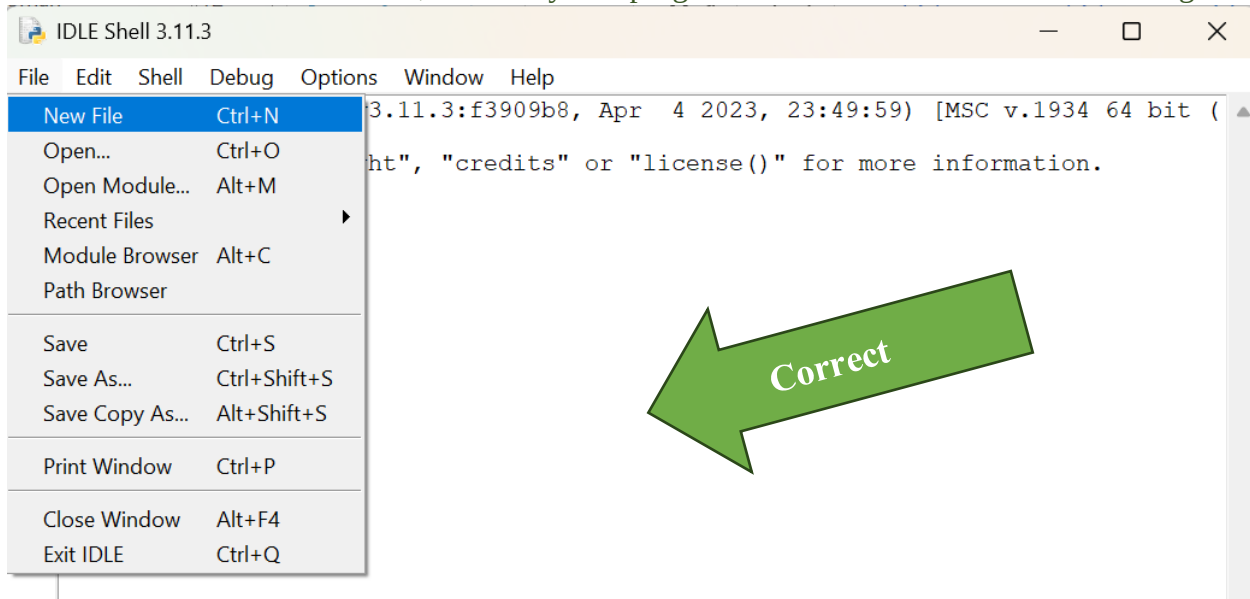


Fig. 2: Creating a new file

- Assign a value of **25** to a variable called **a**. Use the **print()** function to display the value of **a** with a *suitable message*¹.
- Assign a value of **4.2** to a variable called **b**. Print the value of **b** with a *suitable message*¹.
- Calculate the value of **a / b + 5** and assign the result to a variable called **result**. Print the value of **result** with a *suitable message*¹.
- Use the **input()** function to ask the user to enter the name of their favorite fruit. Assign their response to a variable called **fruitname**. Print **fruitname** with a *suitable message*¹.
- Save your file with the name **lab1sol.py**. You can run your program by clicking **Run** → **Run Module** in window containing your **lab1sol.py** file; the output will appear in the IDLE shell, which is also where the user will enter any inputs required by the program.

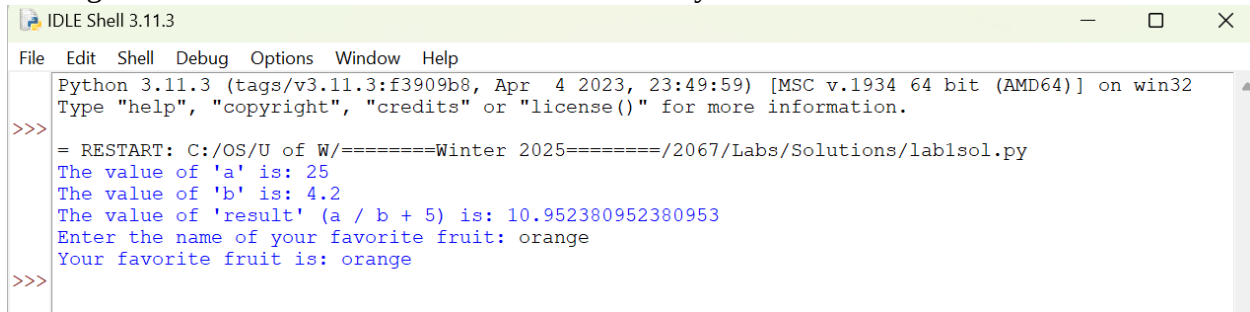
4. Submit your **Python code** (i.e., your **lab1sol.py** file) through Brightspace. The program you wrote in part 3 is the only file you need to submit for this lab².

¹**NOTE:** In this course, the phrase ‘suitable message’ refers to some kind of description that describes what is being printed. For example, just printing the value **22** by itself does not tell the user anything useful; however, printing ‘**x = 22**’ would add some context to what is being printed and so this would qualify as a ‘suitable message’ if the value of x is to be printed.

²**NOTE:** **Do NOT submit a screenshot** or the contents of the IDLE shell displaying your output. **These will NOT be marked** (i.e., you should not submit anything that looks like the image below). You must submit the **lab1sol.py** file.

Below is a **sample** screenshot showing example outputs for Lab 1.

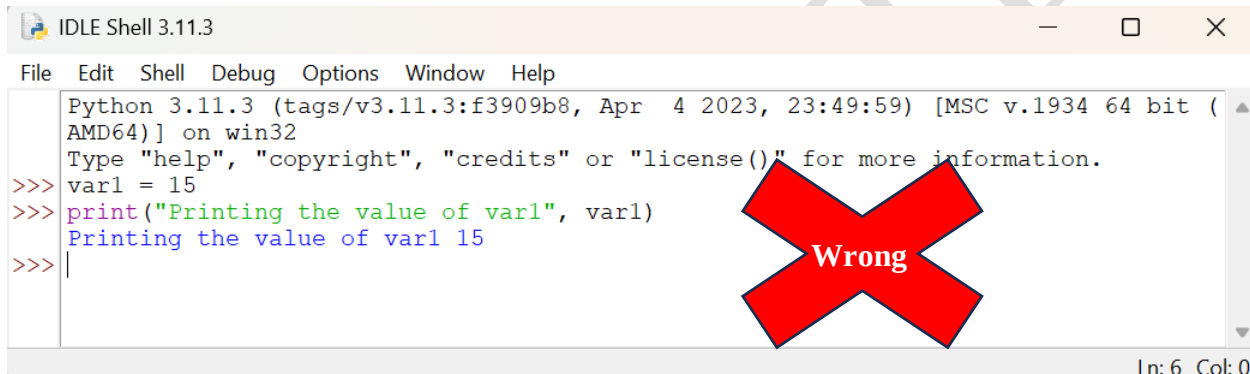
Please note: do not submit a screenshot; only submit your lab1sol.py file. Your suitable message does not need to match the screenshot exactly.



```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/OS/U of W/=====Winter 2025=====2067/Labs/Solutions/lab1sol.py
The value of 'a' is: 25
The value of 'b' is: 4.2
The value of 'result' (a / b + 5) is: 10.952380952380953
Enter the name of your favorite fruit: orange
Your favorite fruit is: orange
>>>
```

Fig. 3. A sample screenshot showing example outputs for Lab 1.

Please note: do not complete this lab using the Python shell.



```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> var1 = 15
>>> print("Printing the value of var1", var1)
Printing the value of var1 15
>>>
```

Fig. 4. Do not complete this lab using the Python shell.

Submission:

1. You will earn a maximum of 100 points (accounts for 2.5%) for submitting your **source code** (*lab1sol.py*) within the specified deadline.
2. **Do not upload your source code to Brightspace as a single zip/rar file.** Such submissions will not be accepted and a mark of **zero** will be assigned.
3. Marks will be **deducted** if a *suitable message* is missing.
4. Late submissions will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. Unlimited resubmissions are allowed. But keep in mind that we will consider the last submission. That means that if you resubmit after the deadline, a penalty will be applied, even if you submitted an earlier version on time.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.

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